

Assignment: Module 2 - The LP Model

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Problem 1: Back Savers

Back Savers manufactures two kinds of backpacks, Collegiate and Mini. Management must decide how many of each to produce weekly to maximize profit while considering limits on nylon fabric, labor hours, and market demand.

a. Clearly define the decision variables

Decision variables are the unknowns we want to solve for the choices managers can make. In this case, management must decide how many units of each type of backpack to produce per week.

- x_C = number of Collegiate backpacks produced per week
- x_M = number of Mini backpacks produced per week

Both variables represent backpack units per week, and together they capture all production decisions available to Back Savers.

b. What is the objective function?

The objective function represents the company's goal. Here, Back Savers wants to maximize **weekly profit**.

- Profit per Collegiate = \$32
- Profit per Mini = \$24

Therefore, the objective function is:

$$\text{Max } Z = 32x_C + 24x_M$$

This formula matters because it links the decision variables directly to the company's profit. By maximizing it, Back Savers ensures resources are used in the most profitable way.

c. What are the constraints?

Constraints are the limits that production must respect. They represent real-world restrictions managers cannot ignore.

1. Fabric availability:

- Total nylon available = 5,000 sq-ft per week.

- Each Collegiate uses 3 sq-ft; each Mini uses 2 sq-ft.
- Constraint: $3xC + 2xM \leq 5000$
- Why it matters: This ensures the company does not use more fabric than it has.

2. Labor availability:

- 35 workers $\times$ 40 hours = 1,400 hours = 84,000 minutes weekly.
- Each Collegiate requires 45 minutes; each Mini requires 40 minutes.
- Constraint: $45xC + 40xM \leq 84,000$
- Why it matters: Labor time is a critical bottleneck resource.

3. Market demand:

- Collegiate: at most 1,000 per week $\rightarrow xC \leq 1000$.
- Mini: at most 1,200 per week $\rightarrow xM \leq 1200$
- Why it matters: There is no reason to produce more than customers will buy.

4. Non-negativity:

- $x_C, x_M \geq 0$
- Why it matters: Producing a negative number of backpacks makes no sense.

d. Full mathematical formulation

$$\text{Max } Z = 32xC + 24xM$$

$$\text{s.t. } 3xC + 2xM \leq 5000 \quad (\text{Fabric})$$

$$45xC + 40xM \leq 84,000 \quad (\text{Labor})$$

$$xC \leq 1000 \quad (\text{Demand - Collegiate})$$

$$xM \leq 1200 \quad (\text{Demand - Mini})$$

$$xC, xM \geq 0 \quad (\text{Non-negativity})$$

Solver result and interpretation

The optimal solution is:

	A	B	C	D	E	F
1		Collegiate (xC)	Mini (xM)			
2	Decision Var	1000	975			
3					0	0
4	Fabric (sq-ft)			Formula (LHS)	4950	5000
5	Labor (mins)			Formula (LHS)	84000	84000
6	Demand xC			Formula (LHS)	1000	1000
7	Demand xM			Formula (LHS)	975	1200
8						
9	Objective			Profit (Z)	55400	

Variable Optimal Value

Collegiate (xC) 1,000

Mini (xM) 975

Profit (Z) **\$55,400**

Using Excel Solver with the Simplex LP method, the optimal solution was found to be producing 1,000 Collegiate backpacks and 975 Mini backpacks. This combination results in a maximum weekly profit of \$55,400. In this solution, the labor constraint (84,000 minutes per week) is exactly binding, meaning that all available labor capacity is fully utilized. The demand constraint for Collegiate backpacks is also binding at 1,000 units, indicating that market demand caps production rather than resources. The fabric constraint is nearly binding ($4,950 \leq 5,000$), which shows fabric is almost completely consumed but still has a small margin left. The demand constraint for Mini backpacks is not binding, since only 975 out of 1,200 possible units are produced.

From a managerial perspective, this solution highlights that labor is the key bottleneck resource. Even though there is some nylon left and Mini backpacks could still be produced up to the demand limit, there are no more labor hours available to support additional production. The results also suggest that Collegiate backpacks should always be prioritized until their maximum demand is met, because they provide higher contribution per unit of labor relative to Minis. Once Collegiate demand is saturated, remaining labor should be allocated to Mini backpacks. This strategy ensures the company maximizes profit while using its labor most efficiently. If Back Savers wants to increase future profits, management should consider expanding labor capacity (such as overtime or hiring), since labor is the true bottleneck while fabric and demand still have some slack.

Problem 2: Weigelt Corporation

Weigelt Corporation can produce Large, Medium, and Small products at three plants. Management must determine how many units of each size to make at each plant to maximize profit. Production is limited by storage space at each plant, overall demand, and a rule requiring all plants to use the same percentage of their capacity.

a. Clearly define the decision variables

Here, decision variables represent units of each product produced at each plant:

- x_{1L}, x_{1M}, x_{1S} = Large, Medium, Small at Plant 1
- x_{2L}, x_{2M}, x_{2S} = Large, Medium, Small at Plant 2
- x_{3L}, x_{3M}, x_{3S} = Large, Medium, Small at Plant 3
- u = percentage of each plant's capacity used ($0 \leq u \leq 1$)

Why it matters: These variables capture the full production decision by size and plant.

b. What is the objective function?

Profits: Large = \$420, Medium = \$360, Small = \$300.

$$\max Z = 420 (x_{1L} + x_{2L} + x_{3L}) + 360 (x_{1M} + x_{2M} + x_{3M}) + 300 (x_{1S} + x_{2S} + x_{3S})$$

Why it matters: This expression converts production decisions into profit, allowing Solver to maximize the company's bottom line.

c. What are the constraints?

1. Equal plant utilization:

- Plant 1: $x_{1L} + x_{1M} + x_{1S} = 750u$
- Plant 2: $x_{2L} + x_{2M} + x_{2S} = 900u$
- Plant 3: $x_{3L} + x_{3M} + x_{3S} = 450u$
- Why it matters: This ensures fairness and equal workload across plants.

2. Storage capacity:

- Plant 1: $20x_{1L} + 15x_{1M} + 12x_{1S} \leq 13,000$
- Plant 2: $20x_{2L} + 15x_{2M} + 12x_{2S} \leq 12,000$
- Plant 3: $20x_{3L} + 15x_{3M} + 12x_{3S} \leq 5,000$
- Why it matters: Storage is physically limited and constrains production.

3. Market demand:

- Large: $x_{1L} + x_{2L} + x_{3L} \leq 900$
- Medium: $x_{1M} + x_{2M} + x_{3M} \leq 1200$
- Small: $x_{1S} + x_{2S} + x_{3S} \leq 750$
- Why it matters: The company cannot sell more than customers want.

4. Non-negativity:

- $x_{ij} \geq 0, 0 \leq u \leq 1$
- Why it matters: Negative production is impossible.

	A	B	C	D	E	F	G	H
	A	B (Large L)	C (Med M)	D (Small S)	E (Total Units)	F	G (Storage LHS) Formula	H (RHS Limit)
1								
2	Utilization u	0.925926						
3	Plant 1	516.6667	177.7778	0	694.4444		13000	13000
4	Plant 2	0	666.6667	166.6667	833.3333		12000	12000
5	Plant 3	0	0	416.6667	416.6667		5000	5000
6								
7	Market Totals	516.6667	844.4444	583.3333				
8	Demand Limit	900	1200	750				
9	Objective					Profit (Z) =	696000	

The Solver solution found an optimal utilization fraction of approximately 0.926, meaning all three plants are producing at about 92.6% of their maximum daily capacity. The resulting optimal production allocation was:

- Plant 1: 516.7 Large, 177.8 Medium, 0 Small (total 694.4 units)
- Plant 2: 0 Large, 666.7 Medium, 166.7 Small (total 833.3 units)
- Plant 3: 0 Large, 0 Medium, 416.7 Small (total 416.7 units)

Utilization fraction (u): 0.926

Profit (Z): **\$696,000**

This plan yields a maximum daily profit of \$696,000. The analysis of constraints reveals that storage space at each plant is the binding factor: Plant 1 fully uses 13,000 sq-ft, Plant 2 uses 12,000 sq-ft, and Plant 3 uses 5,000 sq-ft. None of the market demand limits (900 for Large, 1,200 for Medium, and 750 for Small) are binding, since production stays below those levels. Likewise, plant capacities themselves are not directly binding, since the utilization factor u is slightly below 1; instead, the limiting resource is how much storage each plant has available.

From a decision-making perspective, the results show that the company should treat storage as its most valuable and limiting resource. Production is allocated to maximize profit per unit of storage, which explains why Plant 1 focuses on Large and Medium, Plant 2 balances Medium and Small, and Plant 3 produces only Small products. The equal-utilization requirement is satisfied by ensuring all plants use the same proportion of their capacities. This type of analysis helps managers recognize which constraints drive profitability and which resources to expand in the future. For instance, investing in additional storage capacity could allow greater production and profits without changing labor or plant capacity. In particular, increasing storage at Plants 1 and 2 would be most effective, because both of these plants hit their storage limits even though market demand and plant capacity were not fully utilized.

Summary

Both problems illustrate the importance of formulating business decisions as linear programming models. Problem 1 showed how resource limits (labor) and market demand shape product mix and profitability. Problem 2 demonstrated the impact of special operational rules (equal utilization) and revealed storage as the true bottleneck. In both cases, Solver provided an exact solution that managers can use for planning, resource allocation, and identifying areas for capacity investment.